

BINAY—Lighting Up The Future

Binay
LED
Aviation
Obstruction
LightsBINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon

(As per the International Civil Aviation Organization (ICAO) requirements
for Class A and B structures above 45m)

Available in Low Intensity and Medium Intensity versions (As per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, telecommunication, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Full-and-legal maintenance-free operation
- A long life of 100,000 hours (20 years at 18 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Flexible operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed internally to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT



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ers have provided 'tails'—a combination of copper coils and flexible polymer. The magnetic field creates a vibration in the tail, which allows movement. The electronics and microsensors embedded in the capsule allow the capsule's operator to manipulate the magnetic field that guides the movement of the device.

Robots to increase your efficiency

Ekso Bionics has developed an exoskeleton electronic suit that can help the physically challenged walk. The suit—dubbed Ekso—provides nearly four hours of battery power to its electronic legs. The legs replicate walking by bending the users' knees and lifting their legs with what the company claims is the most natural gait available today.

The suit uses a series of sensors placed around the body and in the accompanying crutches to detect how the user wants it to move. A computer signals the legs to carry out the movements in real time.



Ekso robotic suit

Virtual childbirth to assess risks during delivery

A newly developed computer software—PREDIBIRTH—combined with magnetic resonance imaging (MRI) of a foetus may help physicians better assess a woman's potential for a difficult childbirth, reducing the chance for emergency C-section delivery.

Because a woman's birth canal is curved and not much wider than a foetus head, a baby must move through the canal in a specific sequence of manoeuvres. A failure in the process, such as a head turned the wrong way at the wrong time, can result in dystocia (difficult childbirth). PREDIBIRTH uses 3D reconstructed images of the mother's pelvis and the foetus with possible trajectories of the baby's head through the birth canal to determine the likelihood of a normal delivery.

Using PREDIBIRTH, Dr Olivier Ami, obstetrician, department of radiology, Antoine Bécélères Hospital, Université Paris Sud, France and a team of researchers processed MR images of 24 pregnant women. The results (3D reconstruction of both the pelvis and the foetus along with 72 possible trajectories of the baby's head through the birth canal) were computed retrospectively and measured against delivery outcomes for the 24 women.

"The results in predicting dystocia were highly accurate," Dr Ami said.

Flexible robot

If you thought robots were those hard, rigid and tin-shaped beings, think again. Led by George M. Whitesides, researchers at the Harvard University have designed a rubber robot that can bend and turn like a flexible object.

The 13cm (5-inch) long robot fabricated using soft lithography is inspired by squids, starfish and other animals without skeletons. It has four legs, which can be separately